



COLLEGE OF COMPUTER AND INFORMATION SCIENCE

Academic Year 2024 – 2025

PRACTICUM REPORT DOCUMENT

Inigo Yvan Borja BAYHON

Course Coordinator: Adomar L. Ilao, DIT

Submitted to the Faculty of Mapúa Malayan Colleges Laguna In

Partial Fulfillment of the Requirements for the degree of

Bachelor of Science in Information Technology

OVERVIEW OF THE PRACTICUM ENGAGEMENT



iNeed Solutions Inc., founded in 2024 by Jonarc Cardines, is a technology company based in Makati City, Philippines, that specializes in real-time, data-driven parking and mobility solutions. As a key player in the smart parking systems industry, it serves high-traffic urban locations such as malls, condominiums, hospitals, and hotels. The company offers advanced parking management technologies, including QR/barcode scanning, POS integration, predictive analytics, and cloud-based reporting, all aimed at improving operational efficiency and user convenience.

iNeed Solutions Inc. offers a parking management system that uses real-time data and predictive analytics to optimize space usage and reduce congestion. Through a user-friendly mobile app, drivers can easily find available parking spots and view current rates. The system also supports advanced reservations, allowing users to book spaces ahead of time. This smart solution helps eliminate the hassle of searching for parking, saving both time and resources.

Mission

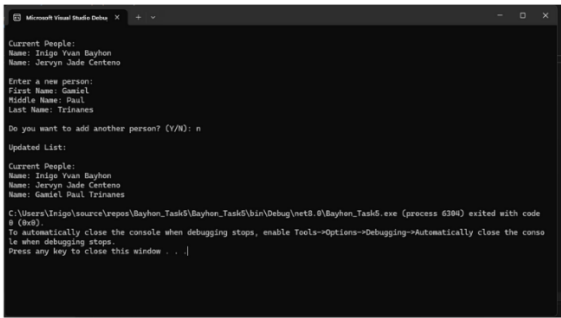
Connect the parkers to thousands of available and accessible parking spaces anytime and anywhere all online.

Vision

Transforming all non-performing land space becomes valuable, useful, and efficiently managed for the customers, owners, cities, and for a national scale.

PRESENTATION OUTPUT

Console App



csv file

A screenshot of a CSV file opened in a spreadsheet application. The application shows a warning: "POSSIBLE DATA LOSS Some features might be lost if you save this workbook in the comma-delimited format". The data is as follows:

	A	B	C	D	E	F	G	H	I
1									
2	Inigo	Ivan	Bayhon						
3	Jervyn	Jade	Centeno						
4	Gamiel	Paul	Trinanes						
5									
6									
7									
8									
9									
10									
11									
12									
13									
14									
15									
16									
17									
18									
19									
20									
21									
22									
23									

Console App

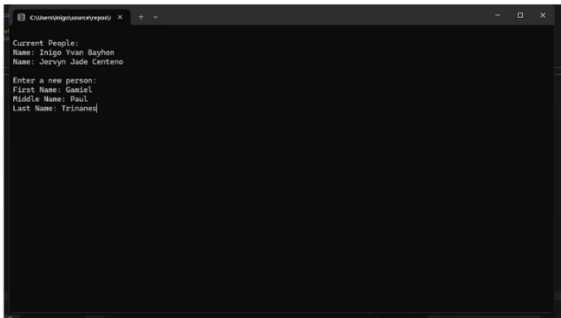


Figure 1. Reading/Writing to an existing .csv File

As shown in Figure 1, a C# console application is used to manage a list of people by displaying current entries and allowing users to add new ones with their first, middle, and last

names. The data is updated in real time based on user input. The person records are also maintained in a spreadsheet format, suggesting integration with an external CSV or Excel file for data storage or synchronization. This setup streamlines data while ensuring that records are consistently saved and accessible for future use.

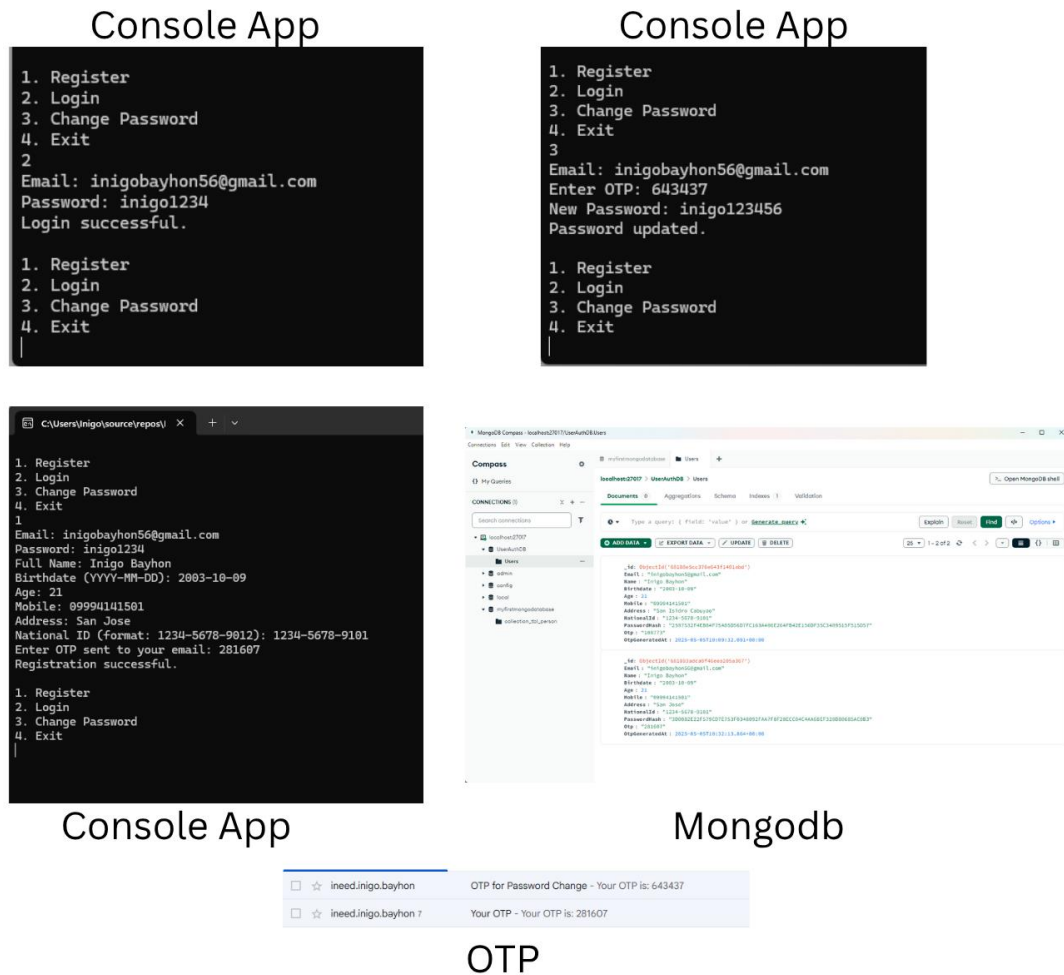


Figure 2. User Registration via Console App and OTP

As shown in Figure 2, this setup showcases a C# console application integrated with MongoDB for user registration, login, and password management. The app allows users to register by entering personal details such as email, name, birth date, age, mobile number, address, and national ID. It includes OTP-based verification for both registration and password change, with OTPs sent to the user's email. Successful actions like login and password updates are confirmed within the console. MongoDB is used as the backend database to store and manage user data

securely. This system demonstrates complete user authentication flow with real-time validation and database integration.

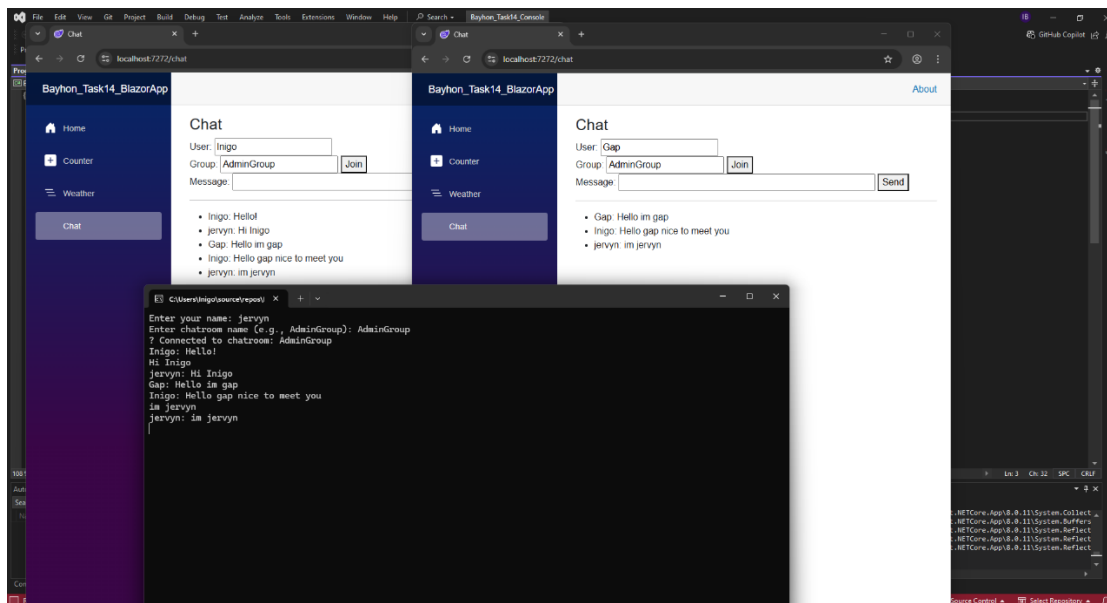


Figure 3. Messaging System using Blazor application and Console application

As shown in Figure 3, this setup demonstrates a real-time chat application built using SignalR, where users from both a Blazor WebApp and a Console App can communicate within the same chatroom. Users enter their names and join a group (e.g., "AdminGroup") to send and receive messages instantly. The Blazor WebApp features a modern UI with a sidebar for navigation, while the Console App shows text-based interaction in the terminal. Messages are synchronized across all clients, showcasing cross-platform communication in real time. This highlights the effectiveness of SignalR in enabling seamless, bi-directional messaging across different application types.

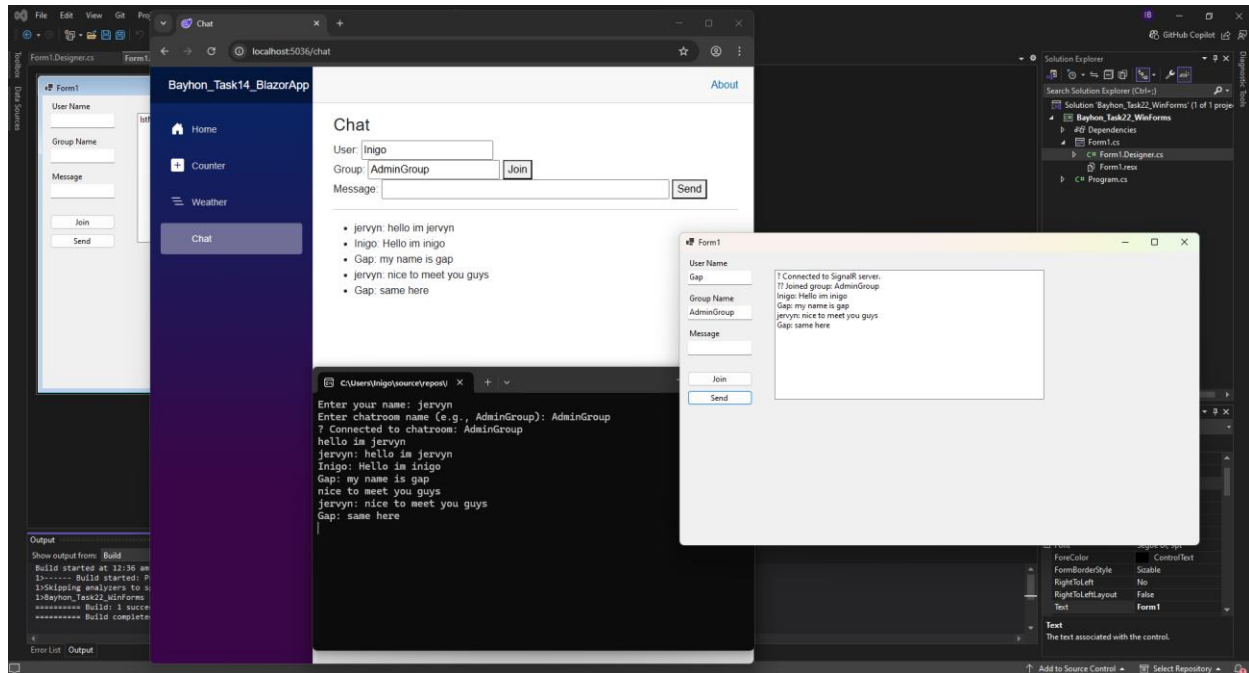


Figure 5. Chat hubs implemented through Blazor application, Console application, and WinForms

As shown in Figure 5, This setup showcases a real-time chat application built with SignalR that connects users across three different platforms: a Blazor WebApp, a WinForms desktop application, and a Console application. Users can join a chat group (e.g., "AdminGroup") by entering their name and group name, and exchange messages that are instantly synchronized across all platforms. The Blazor WebApp features sidebar navigation, while the WinForms app provides a simple form interface, and the Console app offers text-based interaction. This demonstrates how SignalR can effectively enable seamless, cross-platform communication in real-time chat environments. Additionally, it highlights the integration of different .NET technologies within a unified messaging system.

API Endpoint Testing

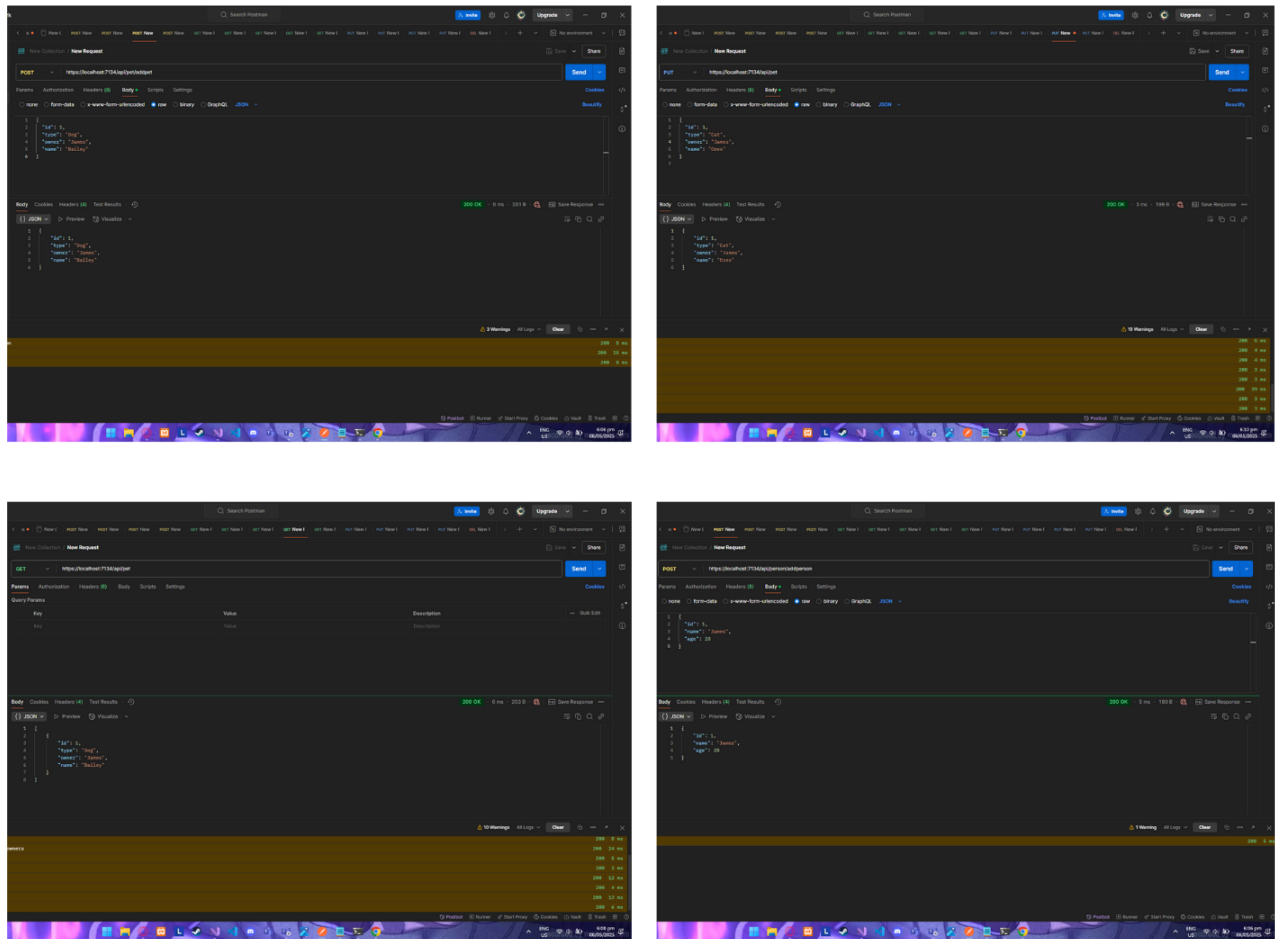


Figure 6. Testing of API Endpoints

As shown in Figure 6, this setup illustrates API endpoint testing using Postman for a user authentication system. Various HTTP requests such as POST and GET are used to test endpoints for functionalities like user registration, login, and password reset. The API responses confirm successful operations with appropriate status codes (e.g., 200 OK) and JSON response bodies. These tests validate that the backend properly handles user data, processes requests accurately, and returns expected outputs, ensuring the reliability and correctness of the API before integration with client applications. This testing approach helps identify potential issues early and ensures smooth communication between the frontend and backend systems.

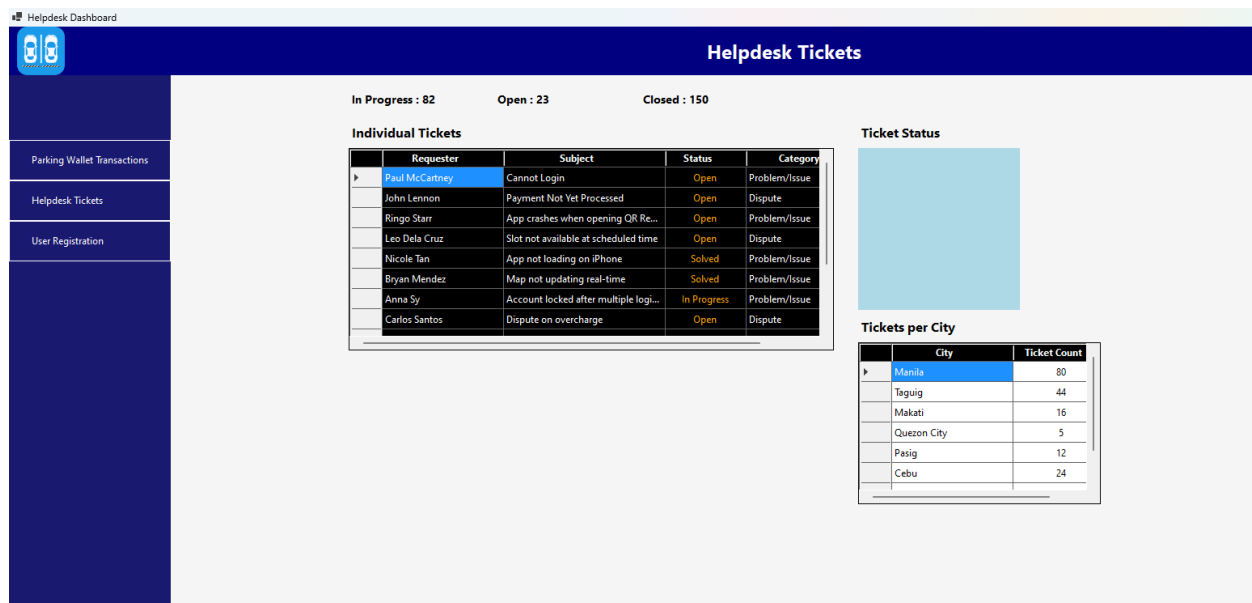


Figure 7. Helpdesk Tickets Dashboard

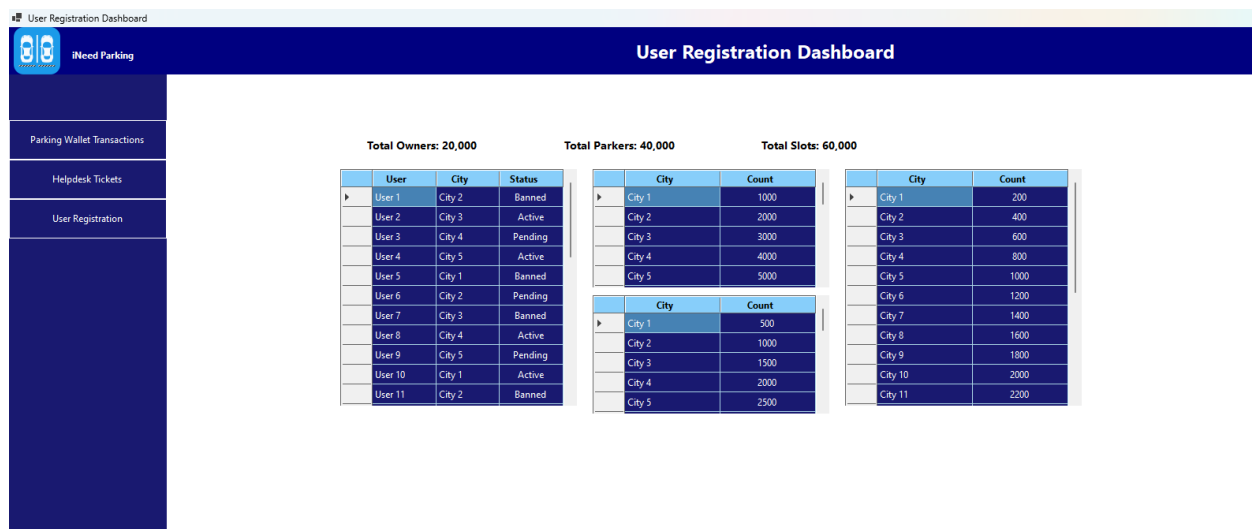


Figure 8. User Registration Dashboard

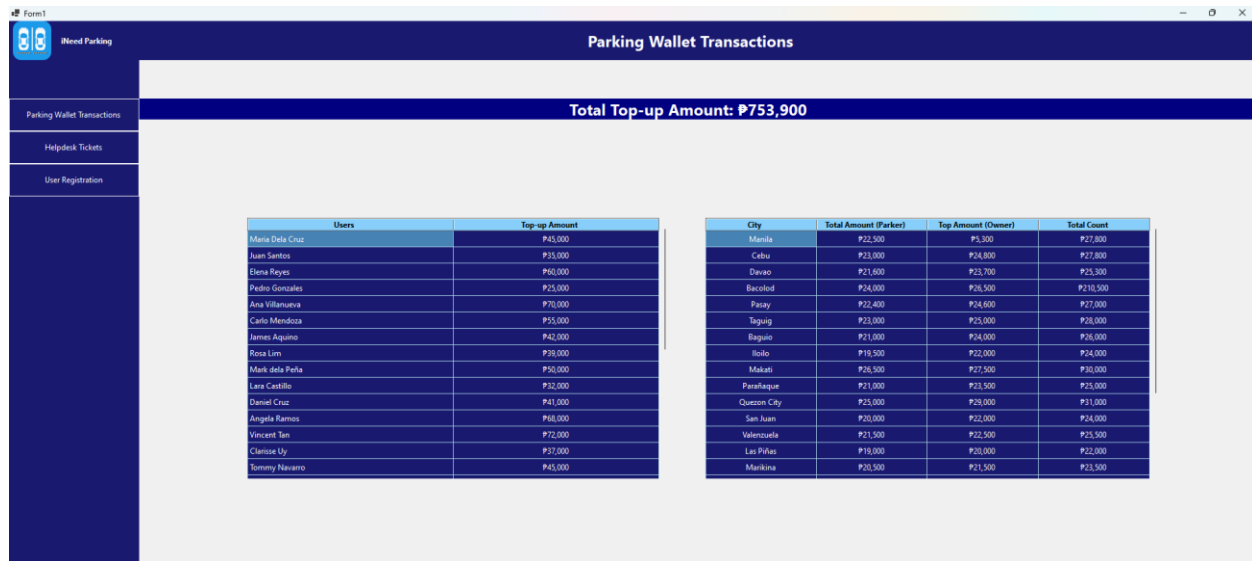


Figure 9. Parking Wallet Transactions

This monitoring system is composed of three main dashboards designed to provide a comprehensive view of operations within the iNeed Parking platform. Each dashboard focuses on a specific aspect of the system: user activity, customer support, and financial transactions. Together, they offer real-time insights that help administrators make informed decisions, ensure service efficiency, and maintain seamless user experience.

As shown in Figure 7, This dashboard provides an overview of the support system's performance by tracking the number and status of submitted tickets. It categorizes tickets based on their issue type and resolution status, helping administrators monitor open, in-progress, and resolved concerns efficiently. A summary of tickets by city is also included to assist in identifying geographic trends in reported issues.

As shown in Figure 8, The User Registration Dashboard presents a summary of registered users within the system, including owners and parkers, along with total parking slots. It also displays the status of user accounts and organizes this information by city. This structure allows for easy monitoring of user distribution, activity levels, and account status across different regions.

As shown in Figure 9, This dashboard displays financial data related to wallet top-ups for both parkers and owners. It includes a breakdown of transaction amounts and distribution by city,

offering a clear view of system-wide monetary activity. This allows administrators to evaluate financial engagement, track usage patterns, and assess regional revenue contributions.

NATURE OF TASKS GIVEN

As a Junior Software Engineer intern, Inigo was assigned to the Software Engineering Department of iNeed Parking, where he collaborated with the development team on various modules across web, desktop, and mobile platforms. He contributed to tasks such as API development, implementation of chat hubs, and system testing. His responsibilities included testing API endpoints for both mobile and web systems, implementing real-time chat hubs using Blazor Server, Console Application, and WinForms, and integrating OTP-based user registration through both the mobile app and console application. He was involved in the design and deployment of messaging systems using Blazor and console applications and helped document technical processes to support future development. These experiences allowed him to develop a deeper understanding of multi-platform development, backend integration, and real-time communication workflows within a functional software ecosystem.

TOTAL HOURS RENDERED

To fulfill the practicum requirement, students must complete a minimum of 486 hours. Out of this total, 406 hours were devoted to applying technical skills, which involved coding assigned modules, exploring new features for learning, producing clean and efficient code, and creating technical documentation for developed functionalities. The other 80 hours were set aside for career growth and skill-building activities, including attending workshops and earning relevant certifications.

SYNTHESIS OF THE PRACTICUM ENGAGEMENT

LEARNINGS

Throughout his practicum, Inigo gained extensive hands on experience in real world software development by contributing to key projects across multiple platforms. He enhanced his skills in C#, .NET, Blazor Server, and WinForms while working on real-time communication features such as chat hubs integrated into Blazor, Console, and WinForms applications using SignalR. He also became proficient in developing and testing API endpoints using Postman and working with MongoDB as the backend for user registration systems. Inigo implemented OTP-based verification workflows on both mobile and console applications, further improving his understanding of secure authentication. Through this practicum, Inigo not only strengthened his technical skillset but also developed a solid foundation in system architecture, documentation practices, and cross-platform application development preparing him for future roles in the software industry.

REALIZATION

During the practicum, Inigo realized that strong collaboration and communication are essential when working on interconnected systems across multiple platforms. Engaging with tasks that involved both frontend and backend components taught him the value of aligning with teammates to ensure smooth integration and consistent outputs. He also came to appreciate the importance of adaptability learning new tools and technologies on the fly to meet project needs. Moreover, the experience reinforced the significance of writing modular and well documented code, which not only supports long-term maintainability but also makes teamwork more efficient and productive.

CONCLUSION

The entire practicum experience proved to be a transformative phase in Inigo's journey as an aspiring software engineer. It provided him with the opportunity to apply his technical knowledge in real-world scenarios, deepen his understanding of system integration, and gain

hands-on experience in multi-platform development. Through this exposure, he improved his skills in coding, debugging, documentation, and collaboration while adapting to industry tools and best practices. More importantly, the experience instilled in him a professional mindset focused on continuous learning, problem-solving, and delivering quality software preparing him to take on more complex challenges in future development projects.

APPENDICES

Appendix A

Competency-Based CV

INIGO YVAN B. BAYHON

San Isidro Heights, San isidro Cabuyao Laguna, Block 2 Lot 23 Phase 1 Extension
inigobayhon@gmail.com
09994141501

PROFILE

My current goal is to secure an On-the-Job Training (OJT) position for my fourth year at university while studying Bachelor of Science in Information Technology. The knowledge I gained from software development classes alongside web and mobile application development and data operations puts me in position to work in teams where I can find solutions and generate new ideas. My professional goals include increasing my knowledge of practical IT operations while bringing substantial value to team projects and advancing my career alongside advancement of innovative technological frameworks.

TECHNICAL SKILLS

- Programming Languages: Python, C#, JavaScript
- Database Management: MySQL, Azure SQL
- Web Development: HTML, CSS, PHP, Bootstrap
- Application & Tools: Visual Studio, XAMPP,
- API Testing & Integration: Postman, REST APIs
- Postman, Microsoft Azure, GitHub
- UI/UX Design Tools: Figma

SOFT SKILLS

- Adaptability and Willingness to Learn
- Dependability and Reliability
- Problem Solving and Logical Thinking
- Eagerness to Improve
- Team Collaboration
- Attention to Detail
- Positive Attitude
- Receptiveness to Feedback

EDUCATION

Bachelor of Science in Information Technology	2021 - Present
Mapua Malayan Colleges Laguna	
Information and communication technology	2019 - 2021
Mapua Malayan Colleges Laguna	
Junior High School	2015 - 2019
Don Bosco College Canlubang	

AWARDS & CERTIFICATIONS

- Dean's List, Within A.Y 2021 - 2024 for 6 terms
- CompTIA IT Fundamentals+ (ITF+) Certification
- Certified Google Cloud Skills Boost: Cloud Computing Fundamentals Finisher
- Certified CodeChum Python Course Graduate
- Philippine Driver's License - DL Code: B

Appendix B

Endorsement Letter



28 April 2025

Dhalia Cardines

Corporate IT and Administration

iNeed Solutions Inc

2nd Floor, Wyce Business Center 9599 Kamagong Barangay San Antonio

Makati City 1203, Philippines

Dear **Ms. Cardines**,

The B.S. in Information Technology program of Mapúa Malayan Colleges Laguna requires their students to undergo Practicum program for a minimum of **486** hours in an academic calendar that will prepare our students to be job-ready after completing their curriculum. This program intends to enable our students to acquire and practice the knowledge and skills expected of a graduate of a B.S. IT program which, in turn, would guarantee continuous supply of IT professionals needed by your company.

We believe that your company can provide the relevant exposure necessary for our students to achieve the intended learning outcomes for the B.S. in Information Technology program. In this regard, I would like to endorse Mr. **Inigo Yvan B. Bayhon** to have his practicum activities in your company as requested.

We thank you for your confidence and trust with us and we look forward to a more meaningful linkage that is mutually beneficial to our students and your company.

With warm regards,

A handwritten signature in black ink, appearing to read 'Adomar L. Ilao'.

ADOMAR L. ILAO, DIT

BSIT Program Chair

College of Computer and Information Science

Mapúa Malayan Colleges Laguna

alilao@mcl.edu.ph

(049) 832-4076

Appendix C

Practicum Confirmation and Acceptance Form

AWATUA SCHOOL

PRACTICUM CONFIRMATION AND ACCEPTANCE FORM

IMPORTANT INFORMATION

- STUDENTS ACCEPTED FOR PRACTICUM IN A HOST COMPANY WILL HAVE TO ACCOMPLISH THIS FORM.
- ASK THE PRACTICUM SUPERVISOR/ COMPANY REPRESENTATIVE TO FILL IN THE DETAILS OF THE TRAINING.
- SUBMIT TO THE PRACTICUM ADVISER/COORDINATOR PRIOR TO THE START OF TRAINING.

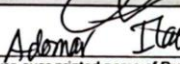
NAME OF STUDENT	INIGO YVAN BORJA BAYHON	STUDENT NUMBER	2021160612
COURSE CODE	IT199F	SY/TERM ENROLLED	2024-2025/3rd Term

This is to certify that Inigo Yvan Borja Bayhon (name of student-trainee) has been accepted for practicum at Ineed Solutions Inc. 2nd Floor, Wyco Business Center 9599 Kamagong Barangay San Antonio Makati City 1203, Philippines (name and address of establishment) and will be attached to the IT department/s for a minimum of, but not limited to 486 hours. Training will commence on April 22, 2025 and is expected to end on July 15, 2025. Attached is the list of requirements.

COMPANY REPRESENTATIVE

 Dhalia T. Cardines	Admin
Signature over Printed Name	Official Designation
Engineering Department	admin@ineed.com.ph / 0917-771-7275
Department	Email and Contact Number/s

NOTED BY

 Ademar Itao	5/2/2025
Signature over printed name of Practicum Coordinator	Date

COPY: (1) STUDENT; (2) HOST COMPANY; (3) PRACTICUM COORDINATOR

FORM OVPA 030B

THIS FORM IS AVAILABLE AT THE OVPA.

Appendix D

Liability Waiver



REVISION NO.: 00
REVISION DATE: May 10, 2016

STUDENT TRAINING AGREEMENT AND LIABILITY WAIVER

IMPORTANT INFORMATION

- THIS FORM IS TO BE ACCOMPLISHED AND SUBMITTED BY STUDENT TRAINEE TO THE PRACTICUM ADVISER BEFORE STARTING THE PRACTICUM.
- READ AND UNDERSTAND THE PROVISIONS OF THIS AGREEMENT AND WAIVER.
- ENSURE THAT ALL SIGNATORIES SIGN THE FORM.

I, Inigo Yvan Borja Bayhon, and a student of MALAYAN COLLEGES LAGUNA (hereinafter referred to as "MCL"), do hereby voluntarily undergo on-the-job training at iNeed Solutions Inc., hereinafter referred to as the "Host Company", located at 2nd Floor, Wynn Business Center BSM Kamagong Surigao San Antonio Makati City 1203, Philippines, under the following terms and conditions:

a. That the practicum training will commence on April 22, 2025 and ends on July 15, 2025 and will have to complete a minimum of 486 hours required for the on-the-job training;

b. That I shall observe proper decorum and act professionally at all times and abide by the Company's rules and regulations and comply with those imposed for the training program, otherwise, I shall be excluded from further participation;

c. That in the course of my training program, I may have access to information which may be of confidential in nature and proprietary to the Company, for which I may be required to execute a confidentiality and non-disclosure agreement as a prerequisite to my participation in the training program;

d. That the time I will spend on the training program in the completion of my on-the-job training requirements will not and should not be interpreted or construed as working hours and should be regarded as non-compensable. Provided that, the Company may, as a unilateral act of liberality or generosity on their part, provide me with meal, travel, transportation allowances, accommodations, etc.;

e. That I fully understand that notwithstanding the allowances enumerated in the preceding section which I may receive, there exists no labor-management and/or employer/employee relationship between me and the Company where I will undergo my training;

f. That I shall exercise due care and diligence in the tasks assigned to me and personally be made answerable for any and all liabilities for damage to property or injury to third person, which may be occasioned by my intentional or negligent acts during the course of my on-the-job training;

g. That I shall likewise hold the Host Company and MCL free and harmless from any and all liability and responsibility for any sickness or injury to myself and third parties and damage to property which I may sustain and/or may occur at any time during the training program, including time spent in traveling to and from any and all premises and locations where I may be required to go to as part of my training program;

h. That the Company reserves the right to discontinue my training on reasonable grounds upon written notice to MCL and myself. Additionally, in the event my training program is discontinued for reasons attributable only to myself, I may be made to reimburse the Host Company for any/all the allowances, stipends, etc., which I may have received from them during and prior to the termination of my training program;

i. That in addition to my liability under section g and for the pre-termination of my training program provided for under section h hereof, I may be subjected further to disciplinary action in accordance with the school's student manual and/or be a ground for disqualification from graduation;

Signed on this 23rd day of April, 2025

Inigo Yvan Borja Bayhon

Signature over printed name of Student Trainee

WITH OUR CONSENT:

Signature over printed name of Parent/Guardian
(for minors only)

NOTED BY:

Adomar Ilao 5/2/25

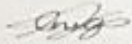
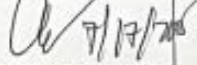
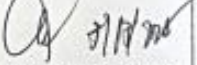
Printed Name and Signature of Practicum Adviser/ Coordinator

Dhalia T. Cardines / iNeed Solution Inc

Printed Name and Signature of Host Company Representative

Appendix E

Training Plan

 MCL Malayan Colleges Laguna A MAJAPAHSCHOOL		REVISION NO. 1 REVISION DATE: May 18, 2018	
TRAINING PLAN			
NAME Inigo Yvan B. Bayhon		COURSE CODE IT199F	
PROGRAM & STUDENT NO. BSIT - 2021160612		COURSE TITLE IT Practicum	
STUDENT OUTCOMES CO1. Identify, analyze, and design business process solutions to the problem faced by the organization. CO2. Apply the different concepts of systems analysis and design, software engineering, database management, and programming courses in the problem-solving process in the organization, and CO3. Acquire new knowledge and experience while in the organization.			
AREAS / PHASES OF TRAINING AND TIME ALLOTMENT A. Demonstration of Technical Skills (Software Development, Research, and Implementation) – 406 hours B. Career Planning and Skills Development – 80 hours			
EVALUATION GUIDELINES & COURSE OUTCOMES			
DEMONSTRATION OF SOFT SKILLS (40%) KEY AREAS COMMUNICATION SKILLS (20%) Relate to co-trainees/supervisors terminologies and rules Recite procedures and instructions needed for the tasks Identify and describe safety signs and symbols Ask critical questions related to the tasks Produce well-written regular and incident reports Prepares and presents reports using Information and Communication Technology (ICT) PROFESSIONAL DEPORTMENT (20%) Observes proper grooming and attire Reports to work regularly on time and as necessary, even beyond prescribed working hour Acts according to the job description given by the company Willing to accept new tasks apart from the usual routine and responsibilities Delivers quality output on time Demonstrates respect for different individuals INITIATIVE (+5%) Volunteers to perform tasks beyond routine tasks		DEMONSTRATION OF TECHNICAL SKILLS (60%) KEY AREAS Demonstration of Technical Skills (40%) - Able to integrate and implement new modules (10%) - Able to deliver modules on time (10%) - Able to write code that is easy to understand and reusable (10%) - Able to identify, analyze, and resolve issues or bugs during development. (5%) - Able to document features, workflows, and technical details clearly and consistently (5%) Career Planning and Skills Development (20%) - Able to understand and apply workplace frameworks and project management principles and adapt to emerging technologies and understand their relevance in the modern workplace (10%) - Able to set career goals and create a personal development plan, and identify personal strengths, demonstrate self-awareness, and develop leadership potential (10%) INITIATIVE (+5%) Volunteers to perform tasks beyond routine tasks	
CONFORME  Inigo Yvan B. Bayhon / 4-25-2025 SIGNATURE OVER PRINTED NAME OF STUDENT / DATE	CONSENT (FOR MINORS ONLY) SIGNATURE OVER PRINTED NAME OF PARENT OR GUARDIAN / DATE	NOTED BY  Jonarc S. Cardines / 2May25 SIGNATURE OVER PRINTED NAME OF PRACTICUM SUPERVISOR / DATE	ENDORSED BY  SIGNATURE OVER PRINTED NAME OF PRACTICUM ADVISOR / DATE
		APPROVED BY  SIGNATURE OVER PRINTED NAME OF PROGRAM HEAD / DATE	
COPY: (1) STUDENT; (2) HOST COMPANY; (3) PRACTICUM COORDINATOR			
FORM OVPAA-0300 THIS FORM IS AVAILABLE AT THE OVPAA.			

Appendix F

Host Evaluation



REVISION NO.: 00
REVISION DATE: May 13, 2016

STUDENT EVALUATION ON PRACTICUM HOST COMPANY AND TRAINING

IMPORTANT INFORMATION

- THIS FORM IS USED TO EVALUATE THE PERFORMANCE OF PRACTICUM HOST COMPANY BY THE STUDENT
- PRACTICUM ADVISER NOTES THE EVALUATION AND DISCUSSES RESULTS WITH THE STUDENT
- NOTED OJT PERFORMANCE EVALUATION REPORT FORMS PART OF THE PRACTICUM REPORT/ PORTFOLIO OF THE STUDENT

NAME OF HOST COMPANY	DEPARTMENT/SECTION/AREA ASSIGNED
iNeed Solutions Inc.	IT
ADDRESS OF COMPANY	2nd Floor, Wyce Business Center 9599 Kamagong Barangay San Antonio Makati City 1203, Philippines

INSTRUCTIONS: Please indicate how much you agree with each statement with 1 being that you strongly disagree and 5 being that you strongly agree.

LEGEND: 5 – Strongly Agree 4 – Agree 3 – Neutral 2 – Disagree 1 – Strongly Disagree NA – Not applicable

PART I: EVALUATION ON PRACTICUM HOST COMPANY

STATEMENTS	RATING (please encircle one)					
1. I was given an orientation about the company rules, regulations, and enough explanation of my practicum assignment at the beginning of the training.	⑤	4	3	2	1	NA
2. The employees I worked with served as resource persons, sharing ideas and materials.	⑤	4	3	2	1	NA
3. The people I worked with were perceptive of my needs.	5	④	3	2	1	NA
4. The practicum supervisor spent time observing my performance.	⑤	4	3	2	1	NA
5. The practicum supervisor provided me with enough constructive criticism.	⑤	4	3	2	1	NA
6. The practicum supervisor sufficiently answered my questions and clarifications.	5	④	3	2	1	NA
7. The practicum supervisor was objective when critiquing my skills.	5	④	3	2	1	NA
8. The demands placed upon me were realistic in this practicum experience.	⑤	4	3	2	1	NA
9. I felt comfortable in my overall relationship with the people in the host company	⑤	4	3	2	1	NA
10. The practicum supervisor was fair in her/his judgment of my skills.	⑤	4	3	2	1	NA
11. I benefited from the supervision provided by the practicum supervisor.	5	④	3	2	1	NA
12. I was given sufficient opportunities for the development of my skills and abilities	⑤	4	3	2	1	NA
13. The practicum supervisor served a good professional model.	⑤	4	3	2	1	NA
14. The company promotes a healthy working environment	⑤	4	3	2	1	NA

ADDITIONAL COMMENTS (STRENGTHS AND AREAS TO IMPROVE)

The practicum host company provided a supportive and professional environment that helped me grow both personally and professionally. My supervisor was approachable and gave constructive feedback that improved my skills. I appreciated the opportunities to apply what I learned in real situations.

PART II: EVALUATION ON PRACTICUM TRAINING

STATEMENTS	RATING (please encircle one)					
1. The training permitted me to generate the minimum number of direct contact hours required within a specified timeframe.	(5)	4	3	2	1	NA
2. The training provided me with experiences that encouraged and developed my interpersonal skills.	(5)	4	3	2	1	NA
3. The training provided me with experiences that encouraged and developed my technical skills.	5	(4)	3	2	1	NA
4. The training provided me with experiences that encouraged and developed my analytical skills.	(5)	4	3	2	1	NA
5. The training provided me with experiences that encouraged and developed my management skills.	(5)	4	3	2	1	NA
6. The training provided me with experiences that encouraged and developed my customer relations skills.	5	(4)	3	2	1	NA
7. Facilities and equipment are adequate and made available for the training	(5)	4	3	2	1	NA
8. Overall, the establishment provided me with a good on-the-job training	(5)	4	3	2	1	NA

ADDITIONAL COMMENTS (STRENGTHS AND AREAS TO IMPROVE)

The training gave me valuable hands-on experience that strengthened both my technical and interpersonal skills. I learned a lot in a supportive and well-equipped environment. The activities provided were relevant and helped build my confidence in real work situations. It would be even better if there were more opportunities for team-based tasks or project involvement.

Inigo Yvan B. Bayhon

SIGNATURE OVER PRINTED NAME OF STUDENT

Appendix G

Weekly Journal – Week 1



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	April 22, 2025 - April 25, 2025	AREA ASSIGNMENT	Software development exercises
TASK	Task 1 - 4	SHIFT/TIME	8am - 5pm

Software Development Activities:

On Wednesday, April 23, 2025, Task 3 was completed, focusing on reading and writing .txt files in a console application. This reinforced file handling in C# and reviewed methods, functions, and console input/output operations. On Thursday, April 24, 2025, Task 2 and Task 4 were completed. Task 2 involved creating a console app to generate triangle patterns, enhancing logic and algorithm design skills, while Task 4 focused on developing a task scheduler, diving into time-based operations and task management in C#.

On Friday, April 25, 2025, discussions centered around classes, lists, methods, and tasks involving CSV files. The focus was on understanding how to manage large datasets and efficiently manipulate them using C# programming features. Several code implementations were shared and refined, enhancing problem-solving skills and programming expertise.

Documentation Activities:

The day began with completing the preparedness checklist on Tuesday, April 22, 2025. This was an important step in ensuring all tasks for the week were clearly outlined and well-organized, setting the tone for a productive work week.

Other IT Related Activities:

On Tuesday, April 22, 2025, the day also involved reviewing the basics of the C# programming language. This review session was crucial for reinforcing foundational knowledge, ensuring that more advanced tasks could be tackled effectively. On Friday, April 25, 2025, the day started with attending the weekly Friday meeting, where project progress was discussed, and upcoming tasks were outlined. This meeting helped prioritize goals and provided the opportunity to address any issues that arose during the week.

TRAINEE'S SIGNATURE

Appendix H

Weekly Journal – Week 2



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	April 28, 2025 - May 2, 2025	AREA ASSIGNMENT	Software development exercises
TASK	Task 5 - 8	SHIFT/TIME	8am - 5pm

Software Development Activities:

On Monday, April 28, 2025, Task 5 was completed, focusing on writing and reading CSV files through a console application. This reinforced skills in file handling and data manipulation in C#. On Tuesday, April 29, 2025, Task 6 was completed, involving sending automated emails via a console application along with a task scheduler, further enhancing the use of automation in C#. A weekly meeting was also attended, where MongoDB was discussed and explored for integration with applications.

On Wednesday, April 30, 2025, Task 7 was completed, which involved editing the code from Task 2 to implement classes and methods for the logical structure of the project. On Thursday, May 1, 2025, Task 8 was initiated, focusing on CRUD operations in a console application connected to a MongoDB database. Finally, on Friday, May 2, 2025, Task 8 was completed, contributing to improved backend logic and external data integration through C# console-based applications.

Documentation Activities:

The day began with completing the preparedness checklist on Monday, April 28, 2025, ensuring that the tasks for the week were organized and that the team was aligned with the project's goals.

Other IT Related Activities:

On Tuesday, April 29, 2025, MongoDB was introduced during the weekly meeting. The day also included further exploration of C# concepts, particularly those related to automated tasks and email handling. Friday, May 2, 2025, started with the weekly team meeting, where project progress was reviewed and upcoming tasks were discussed.

TRAINEE'S SIGNATURE

Appendix I

Weekly Journal – Week 3



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	May 5, 2025 - May 9, 2025	AREA ASSIGNMENT	Software development exercises
TASK	Task 9 - 13	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Monday, May 5 to Friday, May 9, 2025, several key development tasks were completed. On May 5, Task 9 was accomplished, which involved building a user registration console application that utilized OTP for verification and uploaded user data to a MongoDB database. On May 6, Task 10 focused on creating an ASP.NET MVC application that implemented HTTP methods (GET, POST, PUT, DELETE) and tested their functionality using tools like Postman or Insomnia. On May 8, both Task 11 and Task 12 were successfully completed. Task 11 involved building a user registration ASP.NET MVC application that used an API, with OTP functionality and MongoDB integration. Task 12 complemented this by developing an ASP.NET MVC application for APIs and a console app to upload user data to MongoDB. Finally, on May 9, Task 13 was completed, which focused on restructuring an API for a parking management system using principles like single responsibility, dependency injection, clean architecture, and setting up an email sender API.

Documentation Activities:

The week started with completing preparatory work for each task on Monday, May 5, 2025, ensuring all necessary steps were organized and understood for the successful execution of the tasks.

Other IT Related Activities:

On May 6, 2025, additional learning was focused on HTTP methods and their usage in real-world applications, particularly regarding API integrations and testing tools. The weekly meeting was attended, discussing upcoming tasks and areas to improve in system design and data management.

TRAINEE'S SIGNATURE

Appendix J

Weekly Journal – Week 4



REVISION NO.: 00
REVISION DATE: May 10, 2015

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	May 13, 2025 - May 16, 2025	AREA ASSIGNMENT	Software development exercises
TASK	Task 14 - 15	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Tuesday, May 13 to Friday, May 16, 2025, development efforts focused on building real-time communication functionality and exploring cross-platform technologies. On May 13, Task 14 was initiated, which involved developing real-time messaging applications using both a BlazorApp and a ConsoleApp. The goal was to enable users to join a chatroom, specifically "AdminGroup," for the BlazorApp without requiring a login, allowing them to send and receive messages in real-time. The BlazorApp was designed to support multiple users across different tabs. On May 14, Task 14 was completed, and Task 15 (Review) was also accomplished.

On May 15, Task 16 and Task 17 were started, which further expanded on real-time messaging and integration across platforms. Finally, on May 16, the weekly Friday meeting was attended, where discussions were held regarding Flutter development.

Documentation Activities:

The week began with ensuring all necessary preparations were completed on Tuesday, May 13, 2025, setting the stage for the tasks ahead and providing a clear direction for development work.

Other IT Related Activities:

On May 16, the Friday meeting focused on Flutter development, discussing new opportunities and challenges with this cross-platform framework. The meeting provided insights into how it could integrate into the current development projects.

TRAINEE'S SIGNATURE

Appendix K

Weekly Journal – Week 5



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	May 19, 2025 - May 23, 2025	AREA ASSIGNMENT	Software development exercises
TASK	Task 16 - 20	SHIFT/TIME	8am - 5pm

Software Development Activities:

From May 19 to May 23, 2025, key tasks focused on planning, API deployment, and Flutter development. On May 19, Task 16 was completed, involving a conceptual car-parking transaction scenario and defining roles like "Parking Owner" and "Parker." Task 17 was also started. On May 20, Task 17 continued, and Task 19 was completed, which involved hosting the API on IIS using Port 4544 and ensuring it auto-started. On May 21, Task 17 was completed along with Task 18, which involved building a Flutter app with basic UI controls. Task 20 was completed on May 22. Finally, on May 23, Tasks 21 and 22 were started, and the weekly meeting was held to discuss GitHub and group activities. These tasks contributed to a better understanding of backend deployment and cross-platform development using Flutter.

Documentation Activities:

The week began with ensuring all necessary preparations were completed on Monday, May 19, 2025, setting the stage for the tasks ahead and providing a clear direction for development work. Documentation for API hosting and Flutter app development was also updated, ensuring that steps were clear for troubleshooting and future implementation.

Other IT Related Activities:

On May 20, the Friday meeting focused on API deployment and its integration with IIS, offering a chance to discuss the real-world application of deployment techniques. Additionally, discussions on GitHub workflows helped streamline team collaboration for version control and development progress.

TRAINEE'S SIGNATURE

Appendix L

Weekly Journal – Week 6



REVISION NO.: 00
REVISION DATE: May 10, 2015

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	May 26, 2025 - May 30, 2025	AREA ASSIGNMENT	Software development exercises
TASK	Task 21 - 24	SHIFT/TIME	8am - 5pm

Software Development Activities:

From May 26 to May 30, 2025, several tasks focused on video documentation, chat integration, and user registration features. On May 26, a video presentation was recorded for Task 17 and Task 18. On May 27, another video presentation was recorded, this time for Task 19 and Task 20. On May 28, Task 21 was completed, integrating a Flutter mobile app into the real-time chat system from Task 14, demonstrating seamless communication between the Flutter app, web app, and console app. On May 29, Task 23 was completed, which involved implementing JWT authentication for a user registration API, handling unauthorized access, and setting up a MongoDB user with restricted access. Task 24 was also finished, creating a BlazorApp for user registration with features like OTP verification, login/logout, profile editing, and picture uploads. On May 30, documentation and review continued as part of the ongoing development process.

Documentation Activities:

On May 26, the initial video documentation for Task 17 and Task 18 was recorded, setting the foundation for visual communication of the development process. Additional documentation continued on May 30, focusing on recording and reviewing the integration steps for the user registration features and chat system.

Other IT Related Activities:

On May 29, a key focus was on implementing JWT authentication for the user registration API, ensuring that unauthorized access was handled appropriately. This included setting up restricted MongoDB access for secure user management, which is essential for protecting user data and maintaining application integrity.

TRAINEE'S SIGNATURE

Appendix M

Weekly Journal – Week 7



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	June 2, 2025 - June 6, 2025	AREA ASSIGNMENT	Group Project
TASK	Group Project Preparation	SHIFT/TIME	8am - 5pm

Software Development Activities:

From June 2 to June 6, 2025, the focus was on preparing the necessary tools and software for the upcoming group project. Our group was assigned the module for the HQ Monitoring Dashboard, which is responsible for displaying all user data from the app. We coordinated with our trainer to ensure that all software versions were updated and installed correctly. Additionally, we familiarized ourselves with using GitHub, our main platform for uploading and reviewing code contributions from each group. This preparation week was essential in providing us with the foundational knowledge needed for the successful execution of our project. We also practiced using GitHub's collaborative features, such as pull requests and branching, to manage our development workflow effectively. Furthermore, the week gave us the opportunity to troubleshoot and ensure that all necessary environments were set up correctly for smooth collaboration.

Documentation Activities:

This week involved ensuring that all necessary tools and software were correctly set up for the project, and documentation for the HQ Monitoring Dashboard module was started. We also documented our setup process for GitHub, including how to upload and download files safely.

Other IT Related Activities:

We coordinated with our trainer on how to use GitHub effectively, focusing on the process of safely uploading and downloading files. This allowed us to manage our contributions efficiently and ensured we were all on the same page for future collaboration.

TRAINEE'S SIGNATURE

Appendix N

Weekly Journal – Week 8



REVISION NO.: 00
REVISION DATE: May 10, 2015

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	June 9, 2025 - June 13, 2025	AREA ASSIGNMENT	Group Project
TASK	Group Project Preparation	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Monday, June 9 to Friday, June 13, 2025, we worked on creating a user story, use case diagram, UI wireframe, and API endpoints for the HQ Monitoring module. As the group leader appointed by our trainer, I delegated tasks to my groupmates to evenly split the workload. This assignment served as a schematic for what the final system would look like, enhancing our skills in planning, designing, and structuring a new system effectively. The task helped our group improve collaboration and organization, laying a solid foundation for the development process in the coming weeks. We also reviewed different approaches to designing the system architecture, considering both scalability and maintainability. Additionally, I conducted a quick check-in with the team each day to ensure that all tasks were on track and any obstacles were quickly addressed.

Documentation Activities:

This week, I focused on documenting the user story, use case diagram, and the initial UI wireframe, which served as the foundational plan for our module. The documentation process was essential for clearly conveying the purpose and functionality of our module to the entire group.

Other IT Related Activities:

On Thursday, June 12, we explored tools for API integration and discussed their potential impact on the project. This allowed us to align our understanding of the technical stack and prepared us for future implementation tasks.

--

TRAINEE'S SIGNATURE

Appendix O

Weekly Journal – Week 9



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	June 16, 2025 - June 20, 2025	AREA ASSIGNMENT	Software Development
TASK	Planning	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Monday, June 16 to Friday, June 20, 2025, our team focused on planning the design layout of the HQ Monitoring Dashboard. We brainstormed the core functionalities needed to ensure proper connectivity between the dashboard and our backend. This phase allowed us to align our design approach with the technical requirements of the system, ensuring a seamless integration between front-end and back-end components. Additionally, we reviewed potential tools and technologies that would support the overall architecture, ensuring scalability and efficiency for future development.

Documentation Activities:

This week, we documented the initial design concepts, including the layout and functionality of the dashboard, as well as the key API endpoints that would be required for integration with the backend. The documentation also outlined the planned flow of data between the dashboard and backend systems.

Other IT Related Activities:

On Thursday, June 19, we explored different connectivity options for the dashboard and backend to ensure smooth data transfer. This involved researching RESTful APIs and WebSocket implementations to determine the best approach for real-time data updates.

TRAINEE'S SIGNATURE

Appendix P

Weekly Journal – Week 10



REVISION NO.: 00
REVISION DATE: May 10, 2018

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	June 23, 2025 - June 27, 2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Monday, June 23 to Friday, June 27, 2025, our team focused on developing the frontend of the HQ Monitoring Dashboards. We aimed to make the interface closely match our initial schematics to ensure better visibility and design consistency. During this phase, we also worked on preparing the backend to handle and display data retrieved from MongoDB, ensuring smooth integration between the two systems. Additionally, we received feedback on the dashboard's design, which led to revisions that improved the overall layout and flow. This iteration allowed us to enhance user experience and streamline functionality across different sections of the dashboard.

Documentation Activities:

This week, we documented the frontend development process, including UI component structures and the initial integration with the MongoDB database. We also recorded design revisions based on feedback, which will help refine the interface and guide future development stages.

Other IT Related Activities:

On Wednesday, June 25, we explored different methods for optimizing data retrieval from the MongoDB database, ensuring that queries ran efficiently and did not impact the frontend's performance. This allowed us to provide real-time data updates to the dashboard without compromising system speed.

TRAINEE'S SIGNATURE

Appendix Q

Weekly Journal – Week 11



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	June 30, 2025 - July 4, 2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Monday, June 30 to Friday, July 4, 2025, our team continued developing the frontend of the HQ Monitoring Dashboard based on the revisions provided by our trainer. Alongside frontend improvements, we began implementing backend functionalities and started developing the necessary APIs that these functions would connect to. We also conducted initial testing of the APIs using Postman to ensure they were functioning correctly and ready for integration. This week marked the beginning of connecting our system's interface with dynamic data handling, which set the stage for more comprehensive development in the coming weeks.

Documentation Activities:

Throughout the week, we focused on documenting the updated frontend structure, detailing how the dashboard's interface would interact with backend components. Additionally, the API testing procedures were documented to ensure that all team members could follow the same integration protocols.

Other IT Related Activities:

On June 30, 2025, we explored methods for improving API performance and reviewed security protocols to safeguard data. This research was essential in preparing for the full system integration phase, ensuring both functionality and security for the HQ Monitoring Dashboard.

TRAINEE'S SIGNATURE

Appendix R

Weekly Journal – Week 12



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	July 7, 2025 - July 11, 2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Monday, July 14 to Sunday, July 20, 2025, our team continued refining both the frontend and backend components of the HQ Monitoring Dashboard module. During this period, all teams were instructed to push their latest code updates to the shared GitHub repository under the "G9000_API" project for compatibility testing across all modules. In parallel, the WinForms dashboard was placed in a separate repository to keep it distinct from the API code, ensuring that the two components could be developed independently. This decision helped streamline the development process by allowing teams to focus on either the API or the dashboard without overlap. To facilitate a smooth workflow, our trainer provided a step-by-step guide for proper merging and testing procedures. This careful approach ensured the integrity of our codebase, minimized conflicts, and enhanced our understanding of version control and collaborative workflows.

Documentation Activities:

This week, we documented the merging and testing procedures to ensure that all integration steps were properly tracked. The process was critical for maintaining the integrity of the codebase and avoiding conflicts between the API and dashboard.

Other IT Related Activities:

On June 18, we reviewed best practices for using version control effectively, focusing on maintaining separate repositories for the API and the WinForms dashboard. This review was vital in enhancing our collaborative workflow and minimizing potential integration issues.

TRAINEE'S SIGNATURE

Appendix S

Weekly Journal – Week 13



REVISION NO.: 00
REVISION DATE: May 10, 2016

DAILY JOURNAL

IMPORTANT INFORMATION

- INCLUDE TASK ASSIGNMENTS OR MOVEMENTS, REFLECTION ON THE DAY'S NEW LEARNING, ACCOMPLISHMENT, CHALLENGES FACED AND HOW YOU RESPONDED, OBSERVATIONS AND RECOMMENDATIONS ON THE IMPROVEMENT OF SYSTEMS / OPERATION / MANAGEMENT, ETC.
- SCANNED COPIES OF THIS FORM SHALL BE SUBMITTED ON A WEEKLY BASIS THROUGH APPROVED LMS.
- HARD COPIES OF THIS FORM SHOULD BE COMPILED AS PART OF THE STUDENT'S PORTFOLIO.

DATE	July 14, 2025 - July 18, 2025	AREA ASSIGNMENT	Software Development
TASK	Project Development	SHIFT/TIME	8am - 5pm

Software Development Activities:

From Monday, July 14 to Sunday, July 18, 2025, after pushing the final versions of the APIs and dashboards for all teams, the focus shifted to comprehensive testing. Each team was tasked with testing their respective components to ensure compatibility across all modules, checking that the codes and outputs worked seamlessly together. This testing phase was essential for identifying any errors in the code and ensuring there were no duplicate codes across the projects. By carefully validating each module's functionality, we aimed to maintain a clean and efficient codebase, minimizing potential issues during integration. This process also helped enhance our troubleshooting skills and reinforced the importance of thorough testing in collaborative development.

Documentation Activities:

This week, we documented the testing procedures and the results of the tests for each module. The documentation included details of any issues encountered, the steps taken to resolve them, and the final outcomes to ensure transparency and clarity across the team.

Other IT Related Activities:

Throughout the week, we also focused on troubleshooting and refining the testing process, learning how to identify and resolve code conflicts more efficiently. This reinforced the value of effective communication and attention to detail when working within a team environment.

TRAINEE'S SIGNATURE